ALVY RAHMAN MASUM

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Portfolio • GitHub • LinkedIn • LeetCode

Education

United International University (UIU) — Dhaka, Bangladesh
B.Sc. in Computer Science & Engineering, CGPA: 3.22/4.00 — 2022 – Oct 2026
Final Year Design Project: *AI Robotic Waste Management System*

Dhaka Imperial College — Dhaka, Bangladesh
Higher Secondary Certificate (Science), GPA: 5.00/5.00 — 2020

National Ideal School — Dhaka, Bangladesh
Secondary School Certificate (Science), GPA: 5.00/5.00 — 2018

Experience

Oasis Infobyte — Remote
Data Analytics Intern — Oct 2026 – Present

- Applying data-analysis concepts through structured project work, data preparation, exploratory analysis, and analytical problem solving during a one-month internship program.
- Strengthening practical Python-based analysis skills through hands-on assignments and guided project execution.

Basic Japanese Language School / **Miracle Overseas** — Remote
WordPress Developer — Jun 2024 – Jan 2025

- Developed and customized substantial portions of a production institutional website using WordPress, Elementor, responsive layouts, theme customization, plugins, and production content.
- Managed hosting/domain configuration, cross-device fixes, performance improvements, troubleshooting, and iterative UI/content updates.
- Provided approximately six months of post-development maintenance through bug fixing and ongoing content updates.

Projects

FitBondhu – Fitness Guide — Live — GitHub
Next.js, React, TypeScript, Supabase

- Built and deployed a beginner-focused fitness platform with workout guidance, diet planning, account flows, streaks, badges, and reusable React/TypeScript components.

ThiKorben – Local Service Marketplace — GitHub
React Native, Expo, Git/GitHub

- Contributing to worker profiles, community/marketplace flows, job-board/chat, and service screens in a shared feature-branch and pull-request workflow.

UIU Shuttle Management — GitHub
React Native, Expo Location, React Native Maps, Supabase

- Developed driver-to-student shuttle tracking flows with live-map visualization and Supabase-backed application workflows.

Secure Vault Extension — GitHub
Chrome Manifest V3, JavaScript, Web Crypto API

- Built a privacy-focused Chrome extension for encrypted local storage of passwords, API keys, notes, and secrets using AES-256-GCM, PBKDF2, SHA-256 verification, auto-lock, and encrypted backup flows.

Travel & Tourism Management Website — GitHub
PHP, MySQL, HTML/CSS

- Built registration, authentication, booking, hotel, transport, billing, and payment-related workflows in a database-backed multi-page CRUD application.

Research & Engineering

Human-AI Reliance & Trust in Generative AI 2026 – Ongoing
HCI Research, Mixed-Method Study

- Investigating how confidence indicators, verified citations, and uncertainty warnings influence reliance on Generative AI among Bangladeshi users.
- Designed a multi-condition mixed-method study combining structured surveys and semi-structured interviews to examine trust, verification behavior, and appropriate reliance.

AI Robotic Waste Management System – Final Year Project 2025 – 2026 — GitHub
Computer Vision, YOLOv8n, OpenCV, Python, 6-DOF Robotics

- Developed a 6-class computer-vision waste detection pipeline on 596 cleaned images; YOLOv8n baseline achieved 0.666 mAP50, 0.539 mAP50-95, and 3.85 ms/image test inference.
- Integrated real-time visual inference with automated 6-DOF robotic sorting and human-in-the-loop capture of uncertain samples for future retraining.

X-FIBF – Explainable Ransomware Detection 2024 – 2025 — GitHub
Cybersecurity, MHSA-BiLSTM, XAI, File Integrity Monitoring

- Evaluated 62,485 static PE samples and 43,872 API-call sequences in a hybrid detection framework combining static analysis, temporal sequence learning, explainability, and behavioral fingerprinting.
- Achieved 98.28% accuracy, 95.21% balanced accuracy, 98.44% malware recall, 99.22% ROC-AUC, and 1.56% false-negative rate with the MHSA-BiLSTM temporal classifier.

Live LoRa Campus Shuttle Tracker Academic Project
IoT / Embedded Systems, ESP32, GPS, LoRa

- Designed an offline-capable campus shuttle communication concept using LoRa, GPS, and ESP32 for location-aware tracking without continuous internet dependency.

Micro-Mouse Autonomous Maze Solver Academic Project
Embedded Robotics, C++, Flood-Fill, IR Sensing, PID

- Developed autonomous maze-solving logic using flood-fill path planning, IR sensing, and PID-based motion control.

Leadership & Achievements

Turtle Venture Business Competition 2023 Top 5 Finalist
GP Future Nation Maker 2023 Recognized among top national participants
Aspire Leaders Program 2026, Aspire Institute Selected Participant
LeetCode Profile
Regular algorithm and data-structure practice with weekly contest participation and leaderboard appearances.

Skills & Interests

Technical: JavaScript (ES6+), TypeScript, Python, PHP, C++, Java, SQL, HTML5, CSS3; React, Next.js, React Native, Expo, Tailwind CSS; Node.js, Express.js, Laravel, REST APIs; PostgreSQL, MongoDB, MySQL, Supabase
AI / Research: PyTorch, TensorFlow, scikit-learn, OpenCV, Ultralytics YOLO, XGBoost, LightGBM, Computer Vision, HCI Research, IoT / Embedded Systems
Tools: Git, GitHub, Docker, Postman, Linux, npm, VS Code, Vercel, WordPress, Elementor
Certifications: Cisco Networking Essentials; Alison Diploma in Software Testing; GP Tech Professional Series
Languages: Bengali (Native); English (Professional); Japanese (Elementary)

References

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